Historically redlined neighborhoods are more likely to lack greenspace today

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A map of New York City illustrates the overlap between historically redlined neighborhoods and areas lacking green space. Credit: Columbia University's Mailman School of Public Health

Historically redlined neighborhoods are more likely to have a paucity of greenspace today compared to other neighborhoods. The study by researchers at Columbia University Mailman School of Public Health and the University of California, Berkeley and San Francisco,

demonstrates the lasting effects of redlining, a racist mortgage appraisal practice of the 1930s that established and exacerbated racial residential segregation in the United States. Results appear in *Environmental Health Perspectives*.

In the 1930s, the Home Owners' Loan Corporation (HOLC) assigned risk grades to neighborhoods across the country based on racial demographics and other factors. 'Hazardous' areas—often those whose residents included people of color—were outlined in red on HOLC maps. In the decades since, redlined neighborhoods experienced lower levels of private and public investment and have remained segregated.

Mounting evidence indicates that historically redlined neighborhoods contributed to worse health outcomes, as well as elevated exposures to air pollution and other environmental hazards. Areas lacking green space often also have elevated levels of air and noise pollution, as well as higher rates of racial segregation and poverty.

"Though redlining is now outlawed, its effects on urban neighborhoods persist in many ways, including by depriving residents of green space, which is known to promote health and buffer stress," says first author Anthony Nardone, MS, a medical student at the University of California, San Francisco.

"We find lingering effects of racist redlining policies from the 1930s. Future policies should, with the input of local leaders, strive to expand the availability of green space, a health-promoting amenity, in communities of color," adds senior author Joan Casey, Ph.D., assistant professor of environmental health sciences at the Columbia Mailman School.

The researchers examined 72 urban areas in the United States, estimating the association between HOLC grades and greenspace, as

measured by satellite imagery from 2010. They compared greenspace between neighborhoods with different HOLC grades but otherwise similar sociodemographic characteristics according to the 1940s Census so as to isolate the effect of HOLC grades, including redlining. They limited their analysis to neighborhoods that overlapped with 1940 census tract boundaries and considered ecoregion because greenspace will be qualitatively different in, for example, the Southwest compared to the Northeast.

HOLC risk grades were one part of a larger pattern of racist policies. The Federal Housing Administration, tasked with stimulating the private real estate market during the New Deal, would not underwrite insurance on private mortgages that would have desegregated neighborhoods. Similarly, racially restrictive covenants, which were clauses in homeownership deeds, prohibited the future sale of many homes to people of color. More recently—even after the passage of the 1968 Fair Housing Act, which explicitly made redlining illegal—racist banking and real estate practices have persisted and are reflected by the fallout of the subprime mortgage crisis, in which those in communities of color, particularly Black and Latino individuals, were disproportionately targeted with predatory loans and foreclosures by banks.

The authors note that their analysis of satellite imagery does not provide an indication of greenspace quality (for example, green space in places with arid climates may not be reasonable proxies for vicinity to natural environments and their health-related benefits). They also do not distinguish between public and private greenspace or untended forest and manicured parks. In some areas, the presence of green space in the 1930s may also have decreased the likelihood that a neighborhood was redlined.

The researchers say future studies could apply similar methods to analyze metropolitan- and regional-specific associations and assess the

extent to which state, county, or city-level policies may modify observed relationships between HOLC grade and green space. Co-authors include Kara E. Rudolph at Columbia University and Rachel Morello-Frosch at the University of California, Berkeley.

More information: Anthony Nardone et al, Redlines and Greenspace: The Relationship between Historical Redlining and 2010 Greenspace across the United States, *Environmental Health Perspectives* (2021). DOI: 10.1289/EHP7495

Provided by Columbia University's Mailman School of Public Health

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